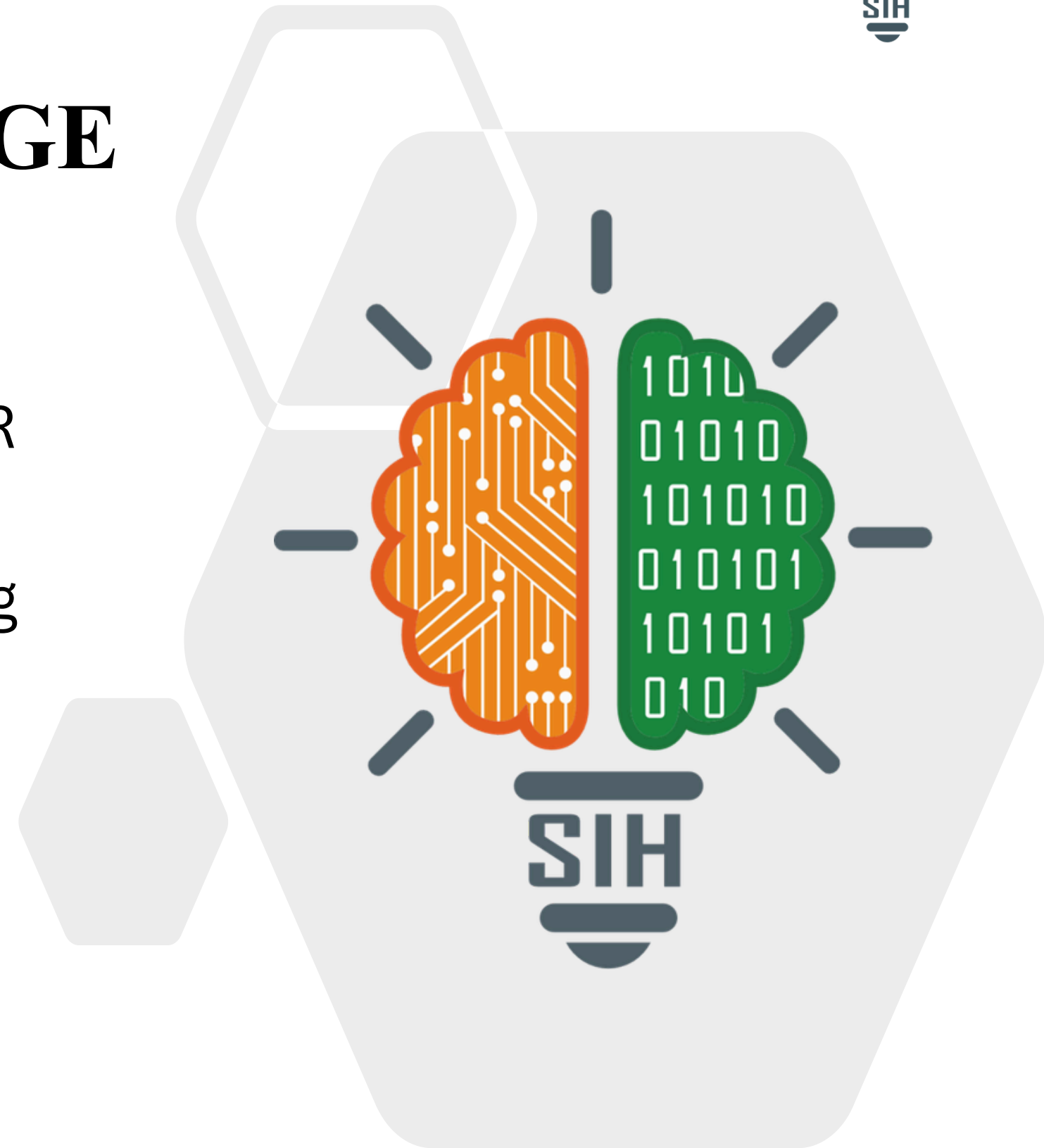


TITLE PAGE

- Problem Statement ID - SIH26177
- Problem Statement Title – A deployable AR powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.
- Theme – Robotics and Drones
- PS Category – Hardware
- Team ID – <Enter Team ID>
- Team Name – Drone Dynamics

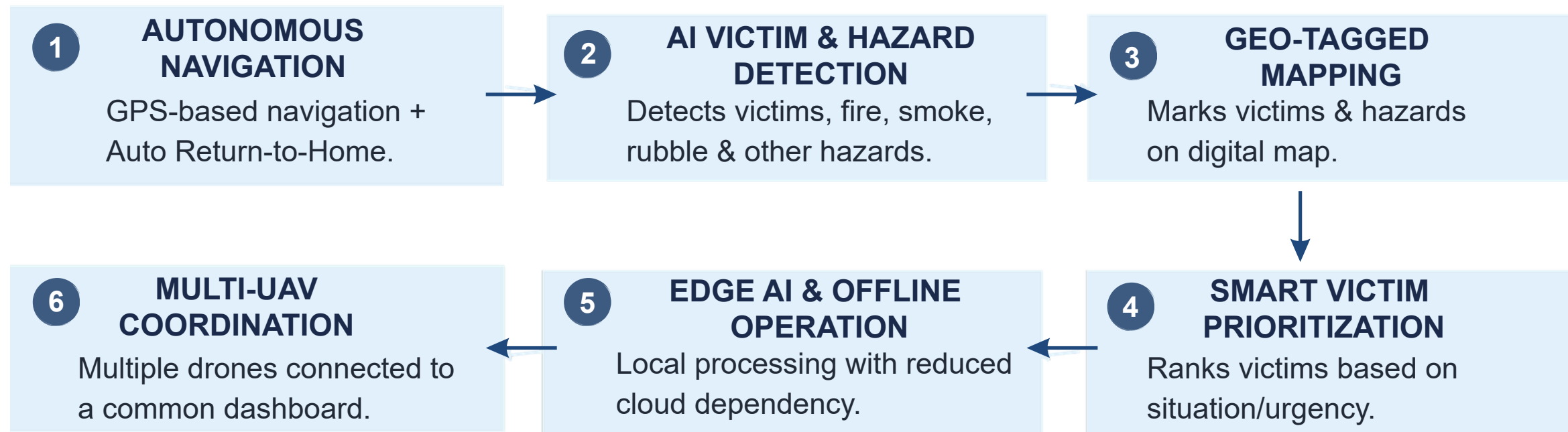




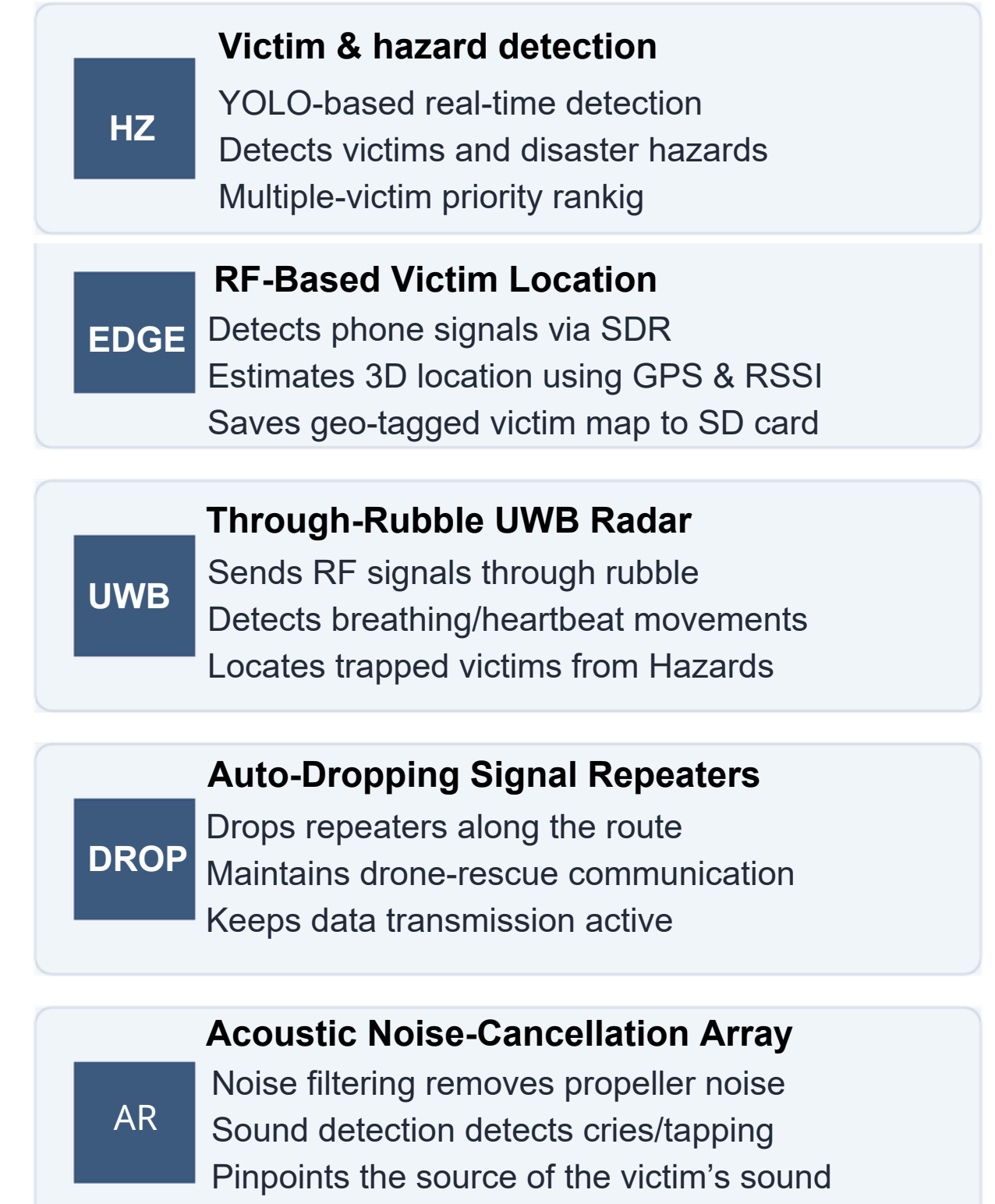
AI - POWERED SEARCH & RESCUE DRONE



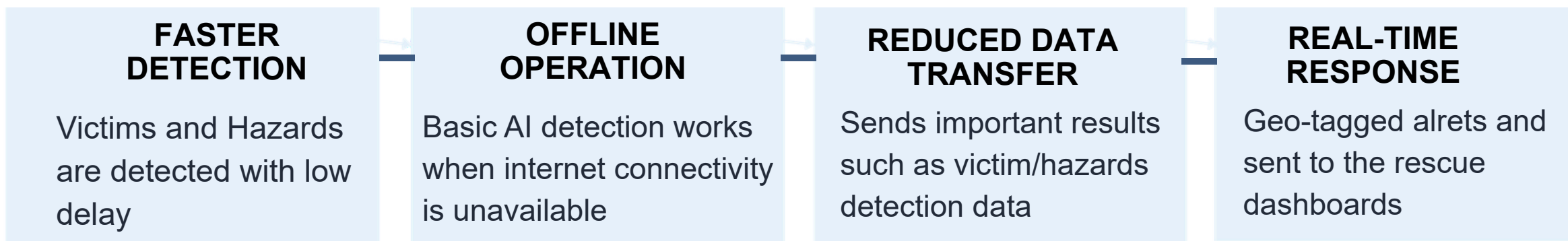
Proposed Solution



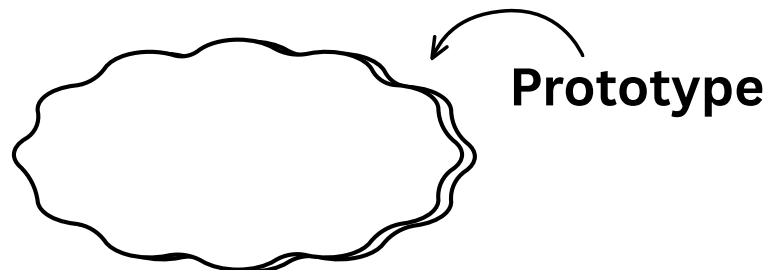
Innovation & Unique Ideas



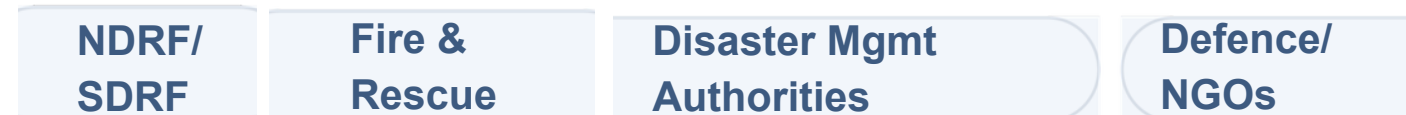
How it Addressed



Innovation: “Our Drone doesn’t just find people - it tells rescuers WHO, WHERE, WHAT RISK, and WHERE TO RESPOND FIRST.”

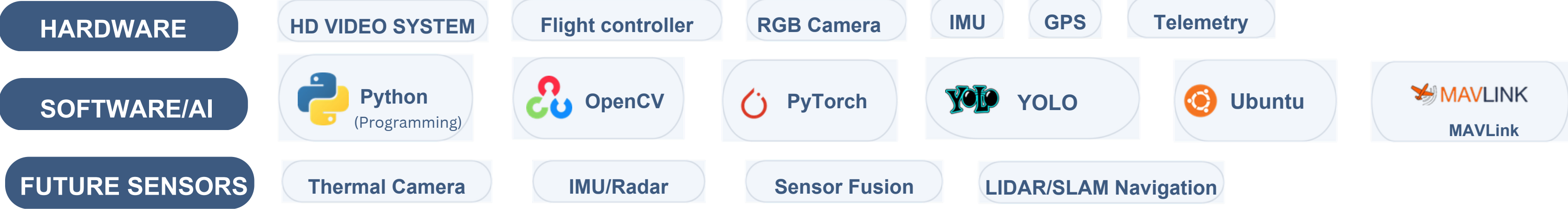


Who can use this

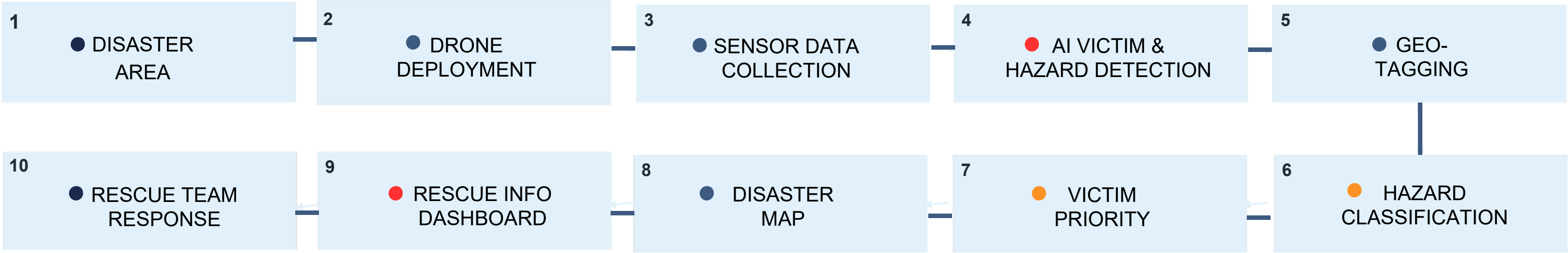




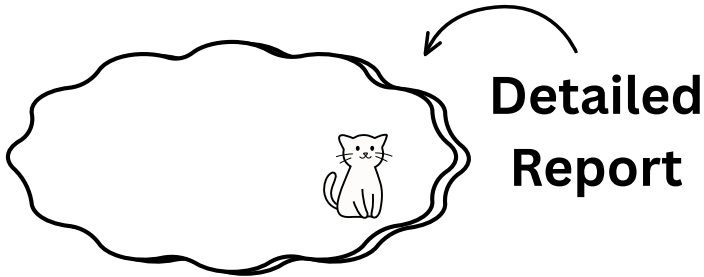
TECHNICAL APPROACH



METHODOLOGY — MISSION FLOW



GPS available → GPS navigation | GPS unavailable → SLAM-based navigation



RESCUE INFO DASHBOARD - LIVE SNAPSHOT

Victim 1 — HIGH	Victim 2 — MEDIUM	Fire Zone — HIGH RISK	Flood Zone — CAUTION	Safe Route — CLEAR
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TECHNICAL

AI/ML(YOLO) for person and hazard detection
Geo-tag mapping for victim and hazard location marking
Directional mic for sos acoustic pulses whistles and screaming
Multi-sensor operating system

FINANCIAL

Open-source tools.
Government grants and schemes(DRDO,ISRO,CSR)
SIH/College/Innovation funding
NGO and industry partnerships

OPERATIONAL

Real time detection and alert generation
Quick deployment in disaster /search areas
Live monitoring through a rescue dashboard
RF-based localisation

SOCIAL

Saves lives and reduces search time
Reduces risk to human rescuers
Improves disaster preparedness and coordination

KEY CHALLENGES & MITIGATION

AUTONOMOUS NAVIGATION

SLAM + obstacle avoidance

SENSOR FUSION

Gradual multi-sensor integration

LOW VISIBILITY

Thermal + LiDAR (future)

COMMS FAILURE

Edge AI + local storage

BATTERY LIMIT

Lightweight, mission planning

DEVELOPMENT ROADMAP

PROTOTYPE

Drone solely doing camera based detection

NAVIGATION & TESTING

YOLO, OPEN-CV VOCABULARY

FIELD TRIALS

With NDRF/SDRF in mock drills

PILOT DEPLOYMENT

Real disaster-zone rollout



IMPACT AND BENEFITS



SPEED

(Time to respond)



ACCURACY

(Geo-Tagged Mapping)



PRIORITIZATION

(Multiple Victim Ranking)



SAFETY

(Reduced Responder Risk)



Potential Impacts on the Target Audience

For Society : Faster rescue & safer communities

For Rescue Teams : Real-time victim/hazard data

For Infrastructure : Rapid damage & hazard mapping

For Governance : Better coordination & data-driven decision making



Supporting Sources

- Private Investment / PPP
- CSR & Industry Contributions
- Disaster Preparedness / Safety Funds



Financial Model

- Government Schemes
- CSR & PPP
- Private & Industry Contributions
- Value rollout / long-term contracts



Field Operations

- Field trials with rescue teams
- Operator training
- State-wise deployment plan

BUSINESS MODEL

Funding Sources for Sustainability



Government Grants:
NDRF/SDRF, Fire & Emergency Services, State Schemes, CSR



CSR & NGO Partnerships:
Industry, Telecom, Disaster Management



Commercial Applications:
Surveillance Mapping, Agriculture, Infrastructure

TARGET AUDIENCE

- NDRF/SDRF
- Fire & Emergency Services
- Police
- Disaster Management Authorities, Municipal/Local Authorities
- NGOs

VALUE ADDED SERVICE

- AI-based person & hazard detection
- Geo-tagged victim/hazard mapping
- Multiple-victim priority dashboard
- Realtime alerts, rescue coordination
- Post-disaster mapping/analysis

Faster search → faster rescue → more lives saved, with safer responders and better disaster preparedness

ALIGNED WITH UN SDGs

SDG 3 — Good Health & Well-Being

SDG 11 — Sustainable Cities & Communities

SDG 13 — Climate Action

PRIMARY USERS

NDRF/SDRF

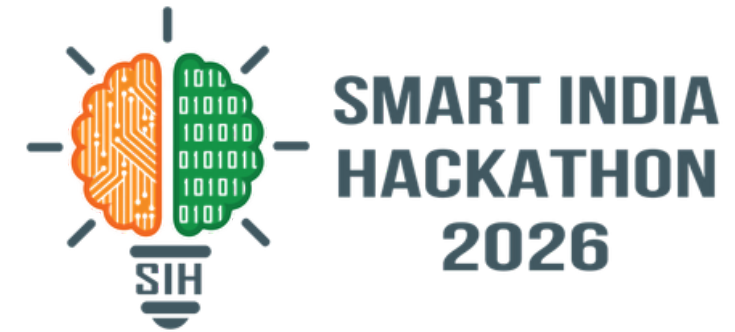
Fire & Rescue

Disaster Mgmt Authorities

Civil Defence/NGOs



RESEARCH AND REFERENCES



1 **PX4 Autopilot Documentation**
<https://docs.px4.io/>

2 **ROS 2 Documentation**
<https://docs.ros.org/>

3 **Ultralytics YOLO Documentation**
<https://docs.ultralytics.com/>

4 **OpenCV Documentation**
<https://docs.opencv.org/5.0/>

5 **Gazebo Simulation Docs**
<https://gazebo.org/>

6 **SIH Problem Statement Portal**
<https://sih.gov.in/>

7 **NDMA Disaster Mgmt Guidelines**
<https://ndma.gov.in/ndma-guidelines>

8 **IEEE Papers — UAV Search & Rescue**
<https://ieeexplore.ieee.org/>

FUTURE RESEARCH DIRECTIONS

Thermal Imaging
Victim detection through smoke, dust & darkness

SLAM Navigation
GPS-denied indoor/
damaged-structure flight

Sensor Fusion
RGB+Thermal+LiDAR+IMU
fusion algorithms

Edge AI Optimization
Lighter models for faster
on-board inference